

Flood Susceptibility Mapping in District Jhang using Remote Sensing and Advanced Machine Learning Techniques

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Abstract— Floods are often considered as a significant natural disaster, primarily due to their devastating effects that result in considerable socio-economic losses. The mapping and assessment of Flood Susceptibility Mapping (FSM) plays a key role in flood prevention and mitigation strategies. Flooding is a natural disaster which is critical issue for many regions, including District Jhang of Pakistan, which faces significant monsoon-related flood risks. In this study, 13 influencing factors of flood occurrence have been assessed by integrating the power of cloud-based computing through Google Earth Engine (GEE) with Geographic Information System (GIS) and programming languages R and Python. For this purpose, Sentinel-2 data were utilized, along with different Machine Learning (ML) and Deep Learning (DL) models. Based on the results from different models, 27.20% to 39% of the study area were projected to be very highly susceptible to flooding. The developed tools and methods in the present research, particularly the FSM visualization app, can be utilized to help decision-makers prepare and act upon the onslaught of a flood event.

Keywords— *Flood susceptibility, Remote Sensing, ML, Google Earth Engine, Jhang, Landsat, Sentinel-2, ANN, RF, SVM, MLP, KNN*

I. INTRODUCTION

Floods accounted for 11.1% of global disaster-related deaths during this period [1]. Several studies indicate that floods impact approximately 200 million people worldwide each year[2]. Pakistan ranks among the most climate change affected countries, and the events of hydrometeorology are very intense here. Floods are a most concerning issue in Pakistan, primarily caused by heavy precipitation during the summer monsoon season and occasionally caused by tropical cyclones movement. Between 1950 and 2010, the regions of southern Punjab, northern Sindh, and northeastern Baluchistan in Pakistan, particularly the Indus Basin, experienced the worst flood impacts[3]. The 2010 flood process started with the unprecedented precipitation in the northern parts of the Indus River and then flowed down to the downstream areas. As such, this imposes a critical need for developing high-performance prediction models that would enable assessing the flood risk and mapping the areas prone to inundation [4].

Spaceborne remotely sensed observations are essential in terms of large-scale flood monitoring and detection applications. For example, the Synthetic Aperture Radar (SAR) data is very useful in the study of flood detection, particularly

during the most difficult weather conditions. The system is independent of external sources of energy, allowing it to capture images at any time[4]. FSM can be defined as the probability that a given area will be flooded, considering various physical, environmental, and meteorological factors. FSM plays a vital role in disaster management, offering valuable insights for preparing and responding to flood events. Accurate FSM maps are vital for mitigating the impacts of flooding and making informed decisions to protect communities and infrastructure. Recent advancements in Remote Sensing (RS) technologies, coupled with ML and DL algorithms, present new opportunities to enhance the accuracy and precision of these assessments[5].

The FSM maps can reduce the damage from floods and helps in, disaster mitigation[6]. According to [7] ML-based models have been found to outperform many other models used in FSM mapping, like the Artificial Neural Networks (ANNs), Random Forest (RF), Support Vector Machine (SVM). The development of Explainable Artificial Intelligence (XAI) has been a salient response to the opacity of state-of-the-art ML models, mostly those referred to as “black boxes”. It uses SHapley Additive explanation’s (SHAP) and interpretation algorithms, trying to make plausible explanations for the model results on the optimal interval of adaptation to urban flooding risk [8].

II. BACKGROUND

Throughout history, people have always faced different forms of natural disasters, but floods have been the most destructive disasters. these floods have distinctly been varying threats in different parts of the world [9]. The climatic conditions range from semi-arid to arid over the major parts of Pakistan and maximum precipitation is observed during the monsoon season (July to September). Floods are common hazard throughout Pakistan, although their nature varies from one region to another[10]. Better ways of predicting floods were encouraged by scientific advancement, which employed sophisticated tools such as Early Warning Systems and Hazard Forecasting Models [11]. The aim of developing the FSM is to strengthen the government and the communities themselves on how to prepare and manage their resources in terms of areas marked as highly susceptible places [12]. Aiming to fill this knowledge gap, this study purposed a novel approach developed Google Earth

Engine (GEE) application for FSM of Jhang which will help to better understand flood prone areas and helps in flood susceptible zones, and any person will easily access this app and seen pervious divesting flood events without any cost. This study used six highly efficient ML and DL algorithms including Multi-Layer Perceptron (MLP), RNN, RF, ANN, SVM and KNN with selected influencing factors using feature selection method, chosen after an extensive literature review and expert opinion, served as the input data for these models. This study integrates cutting-edge algorithms to generate FSM at a sufficient level of accuracy for the study area. This information will be helpful for policymakers in understanding flood risk management and increasing the resiliency of this region.

III. STUDY AREA

Fig. 1 shows the study area Jhang according to geographic coordinates Jhang between 31° N to 73° E. However, district Jhang comprises of four tehsils including Shorkot, Athara Harari, Ahmad PurSial and Jhang. Most of is a flood plain constructed under the depositional work of river Chenab and Jhelum except the western portion. The total area of the study area is 6162.17 km². Heavy rainfall prevails during the different months of the year, with peaks in July, August, and September because of the monsoon season. The study area receives 1242.2 mm/annum rainfall. The average annual temperature in Jhang is around 25°C. The summers are extremely hot, particularly from May to August, when the temperatures rise to 45°C and peak in June. These climatic factors, especially the monsoon rainfall, become particularly important in the flood dynamics of Jhang [13].

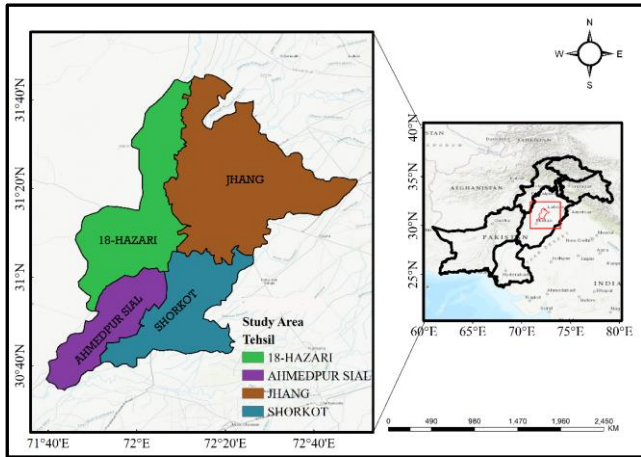


Fig. 1. Map of the study area along with all four tehsils.

IV. MATERIALS AND METHODS

Fig.2. shows methodology that was followed for FSM. ML techniques have been used to map the FSM for the chosen study area. In this study, for the flood event of September 1992, that was the worst flood ever witnessed by Jhelum River, Landsat 5 images from TM sensor have been utilized. Similarly, Landsat 7 images Enhanced Thematic Mapper Plus (ETM+) were used for the 2010 flood and Sentinel 2 images for the 2019 study. Using false color composite imagery and the Modified Normalized Difference Water Index (MNDWI) to identify the flooded areas and to distinguish water bodies. After this, by using this flooded imagery, the data was collected for training

and testing for model training and testing. 70% and 30% of the data were used for training and testing. All 6 models were applied to the dataset with 13 influential variables/factors. The model's accuracy assessment was done using the Receiver Operating Characteristic (ROC) curve, which is highly recommended and widely used[14]. The results from all the six models were used to create FSM. To enhance proper understanding and raise awareness, a GEE app was proposed, which visualizes the historical flood events imagery, influential factors layers, and the FSM maps.

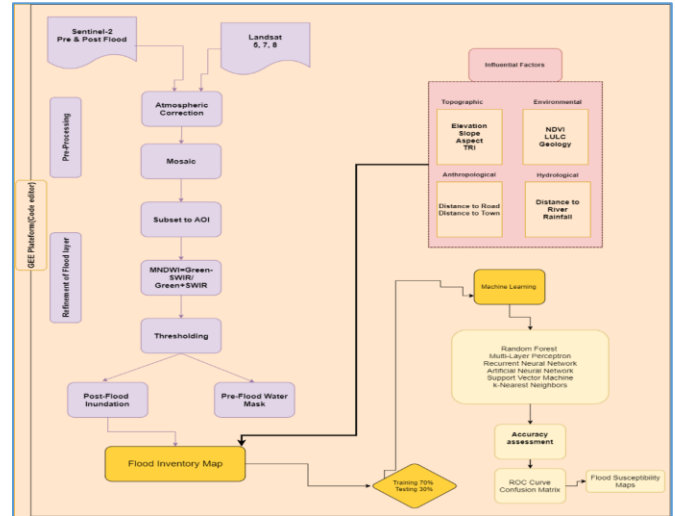


Fig. 2. Complete flow chart methodology of "Flood Susceptibility Mapping , FSM" using sentinel-2 and Landsat imagery. Steps include preprocessing and flood mask generation.

A. Selection of flood influential factors

Fig.3. displays all influential factors used in this study. This study derived several key environmental topographical, anthropological variables using various data sources and geospatial processing techniques were applied. Town and road network extracted using OpenStreetMap (OSM) data (www.openstreetmap.org). HydroSHEDS data were used to extract the water bodies (www.hydrosheds.org), and Euclidean distance calculations were done to calculate the distance from the nearest distance. Elevation data is based on the Advanced Land Observing Satellite Phased Array Type L-band Synthetic Aperture Radar (ALOS PALSAR) dataset, available at (www.eorc.jaxa.jp/ALOS/en/palsar_fnf/fnf_index.html) extracted using GEE that was also used in calculating the Terrain Ruggedness Index (TRI), aspect, and slope. Land Use and Land Cover (LULC) data are based on the Dynamic World Version 4, 2023. Rainfall data are based on the Climate Hazards Group InfraRed Precipitation with Station(/CHIRPS) data available at www.data.chc.ucsb.edu/products/CHIRPS-2.0.

This study calculated other indices, which included Normalized Difference Vegetation Index (NDVI), and for water, it computed the MNDWI using Landsat and Sentinel-2 satellite imagery. Soils information was downloaded from Food and Agriculture Organization (FAO) population data was derived from (<https://sedac.ciesin.columbia.edu/data/set/gpw-v4population-density-rev11>). All these play an important role in water runoff [15].there were 13 conditioning factors were

selected for FSM after evolution their importance like distance to water, rainfall, LULC, elevation, slope, NDVI, and distance to road influential factors were considered for further analyses and model development.

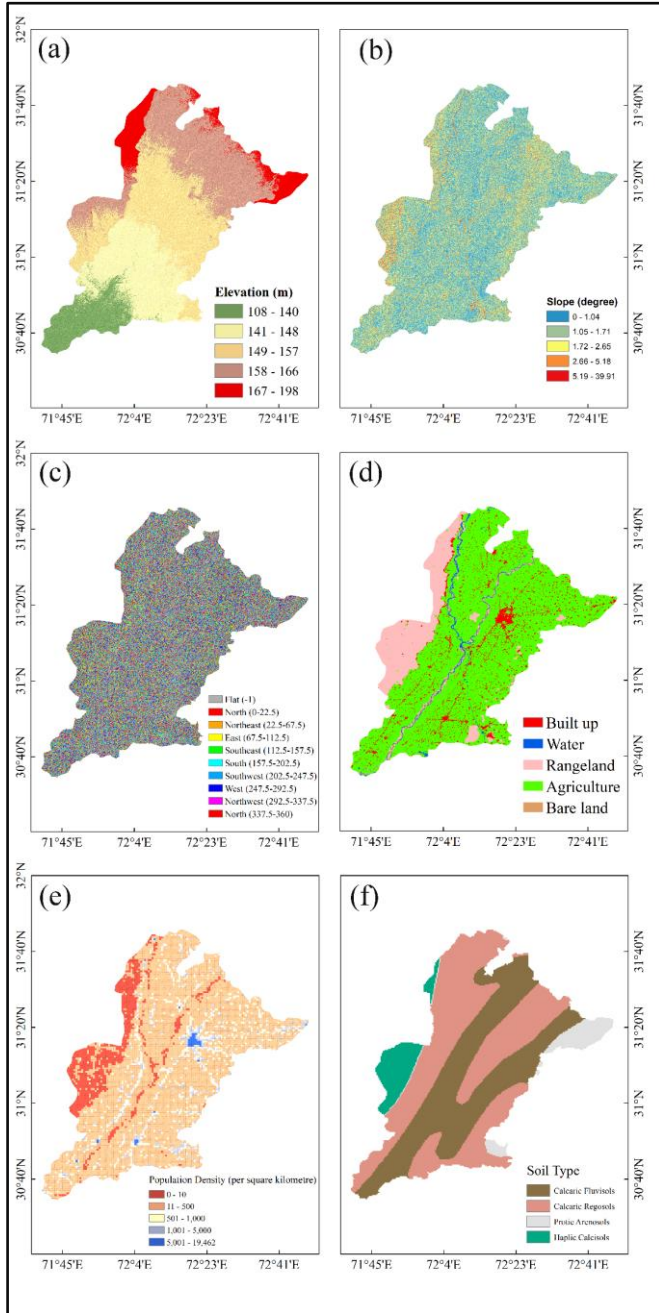


Fig. 3. This figure shows the influential variables affecting flood dynamics: (a) Elevation, (b) Slope, (c) Aspect, (d) LULC, (e) Population Density, (f) Soil

Fig.4. shows the factors including NDVI to distance from town. NDVI is an indicator of vegetation health and abundance. Its value ranges between -1 to +1. A value close to 1 indicates healthy vegetation and value close to -1 indicates no vegetation. NDVI was calculated using defined by the formula (1)[16].

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (1)$$

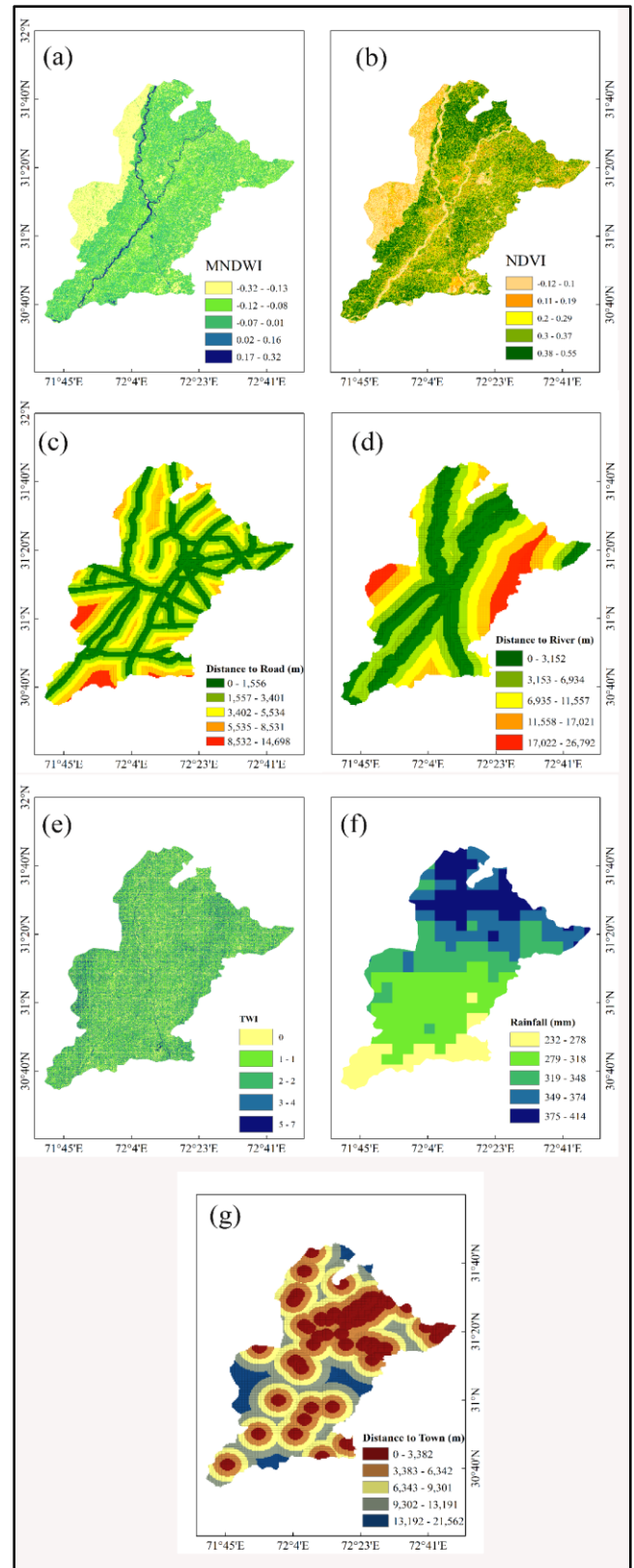


Fig. 4. This figure shows the influential variables affecting flood dynamics: (a) MNDWI, (b) NDVI, (c) Distance to Road, (d) Distance to River, (e) TWI, (f) Rainfall, (g) Distance to Town.

B. Model Development

Model with three layers—input, hidden, and output layers—has been used and found to perform well in FSM [17]. SVM model has been used due to its flexibility of radial basis kernel [18]. RF is used to its capability to handle large data with the help of ensemble of decision trees [19]. DL model Recurrent Neural Network (RNN) utilized because RNN capture periodically some dynamic information in data because of the connection of the hidden layer nodes [20]. According to [21] the most popular and widely used ANN often referred as MLP, made of two layers, input and output, and one or more hidden layers between these two. The hidden layers introduce a network with large ability in modeling complex functions. KNN is a type of algorithm that allows us to classify an unknown entity given the fact that we have data with specified properties (X) and the value of the relationship (Y).

C. Validation Of Models

Fig.5. displays the confusion matrix which is used to determine correlation between variables and target (flood) and feature importance shows which variable is contributing more and next all models ROC curves show performance of each model. This study validated six FSM models by ROC curve analysis based both on validation and training points shown in the accuracy of the models is determined by the Area Under the ROC curve. Curves nearest to the coordinate ($x = 0, y = 1$) (top left corner) and which are above the line of equality are better at predicting a correct result of given dataset. Area Under the Curve (AUC) or c-statistic measures the general ability of a test to distinguish 2 outcomes. The scores for AUC range between 0.5 and 1.0. An AUC value of 1.0 indicates perfect result. A test could be considered excellent 0.9–0.99, good, 0.8–0.89, acceptable 0.7–0.79 while anything less than 0.7 would be nothing to use in this study [22]. The AUC values obtained from the ROC curve analysis in the training dataset are KNN returned the highest AUC of 0.99, followed by ANN with an AUC of 0.98, RF with an AUC of 0.98, MLP returned an AUC of 0.98, RNN returned an AUC of 0.99 and SVM returned an AUC of 0.95. according to AUC all model exhibits perfect prediction results. The correlation matrix showed relationship of all influential variables with target variables distance to water show negative 0.62 correlation coefficient with flood it means their areas are near the water are highly susceptible.

Lower slopes -0.47 showing also susceptibility and same aspect with 0 are highly susceptible. The explainability of ML models comes from XAI which tells about model performance and feature importance and helps to understand which factors significantly impact model predictions. This study used SHAP to identify key influences on model performance like distance from river, rainfall showed more influence than others. As the distance of river channel will close its hits more in these areas. Rainfall is another important factor that plays a crucial role in flooding. The third factor was DEM (Digital Elevation Model) also called elevation. Areas having low elevation are more susceptible compared to areas with high elevation. LULC also plays an important role in FSM the mostly the built up and agricultural areas showed high susceptibility because both comes in human loss.

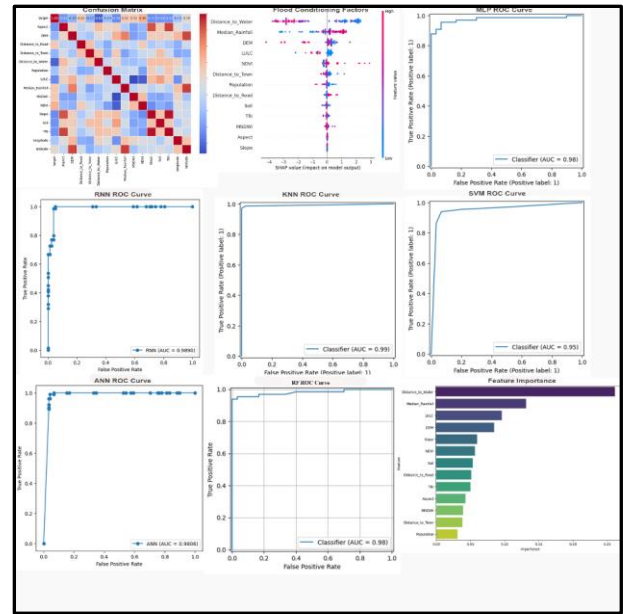


Fig. 5. Performance and accuracy matrix included confusion matrix, “Receiver Operating Characteristic ,ROC “ curve, feature importance.

V. RESULTS

Fig.6. displays the FSM results of all models. For the study of the flood susceptibility analysis using RNN, ANN, RF, MLP, SVM, and KNN models clearly showed remarkable spatial variation across the study area. The susceptibility maps of the region were classified into Very Low, Low, Medium, High, and Very High. The susceptibility indexes were then interpolated using Inverse Distance Weighted (IDW) interpolation on the entire region thus providing the final FSM. According to the RNN and KNN results, the highest proportions for very high susceptibility are shown, especially within the central areas of study area. These areas have low elevation areas. These high susceptibility areas are mostly along riverbeds with lower elevations and slopes. When flood occurs, low elevated regions serve as a natural flood basin bed water mostly accumulated and stored and cause the most damage. The RF model represents a more detailed and less fragmented risk distribution compared to that of the RNN model. the RF map reveals smaller, distinct clusters.

This detailed distribution suggests that the RF model may capture complex patterns between elevation, proximity to water bodies, and other environmental factors more effectively than other models. From the bar graph shown in bottom, it can be clearly noted that the MLP model predicts the largest area under medium susceptibility, the MLP model shows that while not all regions are at likely risk of severe flooding, much of the area could still face flood impacts, especially under intense rainfall conditions while RNN, ANN, and RF models highlight higher risks. In this study, it should be noted that 29% to 39% of the area is highly susceptible to flooding. This study is useful as it suggests that while some areas may not face extreme flooding risks, they are still susceptible and require attention, especially during periods of heavy rainfall or other exacerbating conditions.

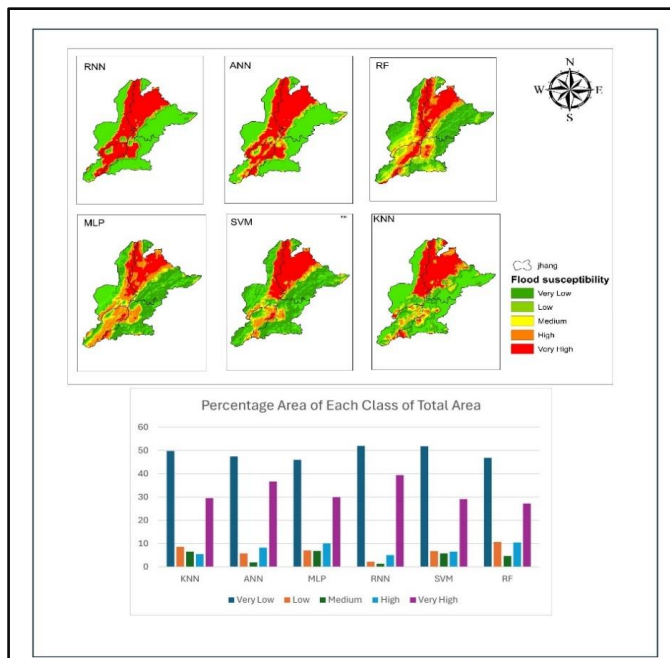


Fig. 6. “Flood Susceptibility Mapping ,FSM” generated maps using different models: RNN, ANN, RF, MLP, SVM, and KNN. Each map shows areas classified by flood susceptibility levels, with red color indicating higher risk zones and green shows low risk zones. The bottom panel illustrates the total susceptible area percentage for each susceptibility class, enabling comparison across the model predictions.

VI. WEB APPLICATION DEVELOPMENT

Fig.7. displays the GEE app interface where panel (a) displays NDVI of study area highlighting vegetation health and density. After selection from dropdown menu, areas near the river show healthy and dense vegetation and red colored areas highlighting there is no vegetation. And panel (b) displays the flood 1992 false color imagery where the dark blue color indicates effected region, areas near the river shows effected areas. To enhance the decision-making process and make a complete flood risk assessment tool, we developed a cloud-based application using GEE. The app integrates all influential variables user can easily select and visualize next is FSM in which user can easily select layer of FSM, and this study also added historical flood Maps for ease of access to make better decisions.

The main objectives are to provide easily accessible, real-time visualization and analysis of the flood risk in the study area to the stakeholders. Layers, such as these data, are pulled together in an app, GEE offers a way to enable knowledgeable decisions for disaster management, urban planning, and resource allocation. Dynamic maps, comparison of different susceptibility models, and historical flood patterns can all be used to improve preparation and mitigation of potential flood impacts. The application provides a user-friendly interface and real-time data, which aids in the implementation of more effective flood risk management practices. This helps foster community resilience through the knowledge and tools provided to the local populations to respond to flooding events. In essence, the initiative aims to enhance safety, protect livelihoods, and promote sustainable development in flood prone areas. Link of application is given below.

<https://syedadnanshahn.users.earthengine.app/view/flood-susceptibility-jhang>.

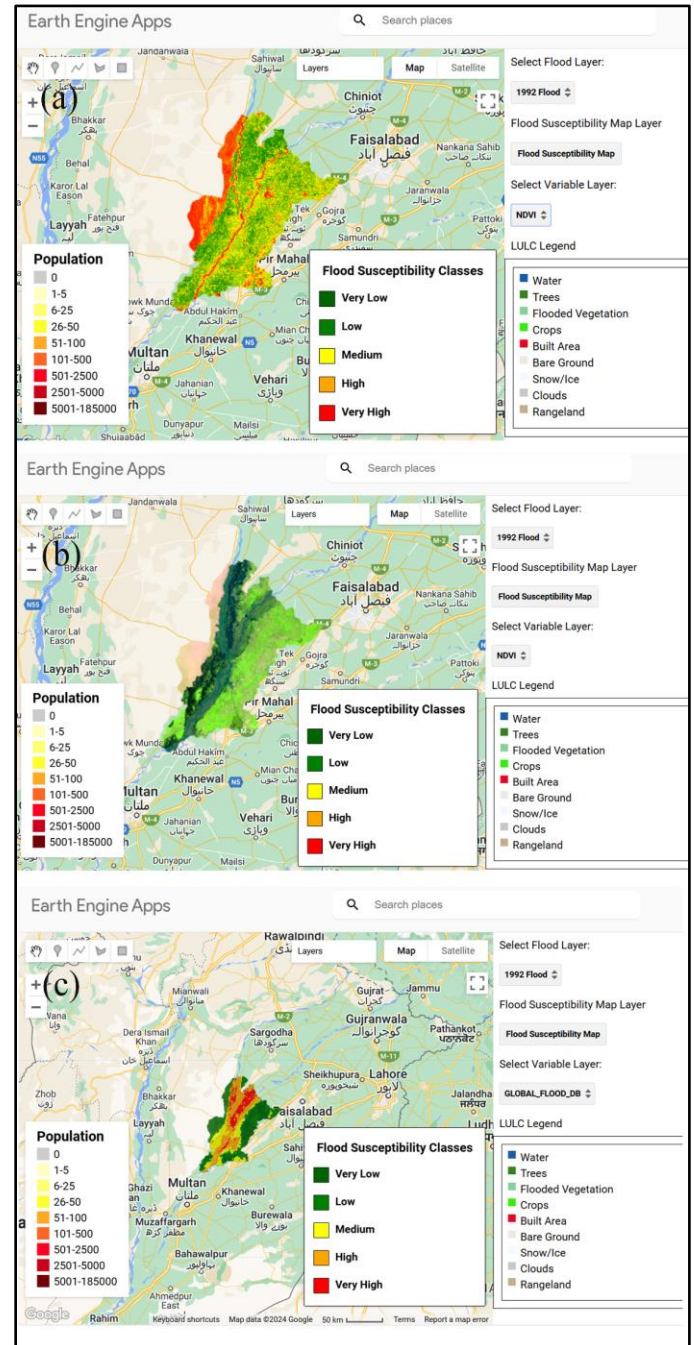


Fig. 7. GEE application interface showing FSM maps and layers. Panel (a) is the NDVI layer, showing vegetation density in the study area. Panel (b) presents the flood extent during the 1992 flood event, and (c) shows the FSM map with Classes very low to high.

VII. CONCLUSION

There are several key measures toward developing management strategies and allocation of resources with respect to such flood hazardous phenomena in the future are the identification and zoning of flood prone areas of areas of high susceptibility specially areas near river having low elevation.

While ML techniques have been applied universally for prediction of floods, it is still a challenging task to produce a reliable FSM map. In this research, we applied six ML methods and checked their predictive performance in Jhang, Pakistan. In FSM, 13 flood influencing factors were used. The result showed that, compared to the other five methods, KNN, RNN, RF were dominant for overall training and validation performance. Unfortunately, the development of reliable hydraulic models is not possible due to the lack of river gauges with adequate observations within the watershed. Under this study, the five ML models used here could be an effective alternate for FSM mapping.

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